# Occupancy likelihood

## Parametrisation

This is a specialized likelihood to for occupancy models.

### Details

An observation is an vector $y = (y_1, \ldots, y_m)$ of binary observations, each depending on spesific covariates, with additional zero-inflation. If there fewer than $m$ observations, like $m' < m$, then "observations not observed" must be set to `NA`. The likelihood for one observation(-vector) is

$$f(y) = \phi \prod_{i=1}^{m} p_i^{y_i} (1 - p_i)^{1-y_i} + (1 - \phi) 1_{[y_i = 0, \forall i]}$$

with the convension that if $y_i$=`NA`, the contribution from $y_i$ is ignored. Further,

$$\text{logit}(p_i) = x_i^T \beta$$

and $x_i = (x_{i1}, \ldots, x_{ik})$ are the $k > 0$ covariate associated to $y_i$, and $\beta = (\beta_1, \ldots, \beta_k)$ are the regression coefficients. The linear predictor from the formula $\eta$, goes into $\phi$, as

$$\text{logit}(\phi) = \eta$$

## Link-function

The link-function for the $p_i$-model is given by argument `link.simple` in the `control.family`-argument. The link-function for the $\phi$-model is given as normal. Both defaults to the logit-link.

## Hyperparameters

The regression coefficients $\beta$ are treated as hyperparameters, and $k$ is maximum 10. An intercept must be defined manually by adding a constant covariate vector.

## Specification

- `family="occupancy"`
- Required arguments: A matrix $Y$ with the observations and a matrix $X$ with the covariates.

The matrix $Y$ is $n \times m$, where $m$ is then the *maximum* number of observations over all locations. If there fewer than $m$ observations in one location, then `NA` must be added to reach $m$.

The matrix $X$ stored the covariates. Since there are $k$ covariate for each observation, then the dimension of $X$ is $n \times mk$. For the $i$th observation(-vector), then the $i$th row of $X$ is $(x_{i1}, x_{i2}, \ldots, x_{im})$, where $x_{ij}$ is the covariate vector for $j$th observation at location $i$. In a single observation is `NA`, the corresponding covariate(-vectors) is not used (but it still needs to be given).

Here is a simple example with $n = 5$ locations, maximum $m = 3$ observations, and $k = 2$ covariates.

```
> Y
     [,1] [,2] [,3]
[1,]    1    1   NA
[2,]    0    1    0
[3,]    0    0    0
[4,]    0    1    0
[5,]   NA    0    0
> round(dig=3,X)
     [,1]   [,2] [,3]   [,4] [,5]   [,6]
[1,]    1 -0.129    1 -0.248   NA     NA
[2,]    1 -0.030    1  0.151    1  0.100
[3,]    1 -0.148    1 -0.061    1  0.023
[4,]    1 -0.055    1  0.081    1 -0.112
[5,]   NA     NA    1  0.164    1  0.027
```

For $Y[1,1]$ the covariates are $(1, -0.129)$, for $Y[1,2]$ the covariates are $(1, -0.248)$, for $Y[1,3]$ is `NA` hence not used, for $Y[2,1]$ the covariates are $(1, -0.030)$, etc. We pass both $Y$ and $X$ in the `inla.mdata()` in the formula, as

```
inla.mdata(Y,X) ~ ...
```

### Hyperparameter spesification and default values

**doc** `Occupancy likelihood`

**hyper**

**theta1**

**hyperid** `56601`
**name** `beta1`
**short.name** `beta1`
**output.name** `beta1 for occupancy observations`
**output.name.intern** `beta1 for occupancy observations`
**initial** `-2`
**fixed** `FALSE`
**prior** `normal`
**param** `-2 10`
**to.theta** `function(x) x`
**from.theta** `function(x) x`

**theta2**

**hyperid** `56602`
**name** `beta2`
**short.name** `beta2`
**output.name** `beta2 for occupancy observations`
**output.name.intern** `beta2 for occupancy observations`
**initial** `0`
**fixed** `FALSE`
**prior** `normal`
**param** `0 10`
**to.theta** `function(x) x`

**from.theta** `function(x) x`

**theta3**

**hyperid** `56603`
**name** `beta3`
**short.name** `beta3`
**output.name** `beta3 for occupancy observations`
**output.name.intern** `beta3 for occupancy observations`
**initial** `0`
**fixed** `FALSE`
**prior** `normal`
**param** `0 10`
**to.theta** `function(x) x`
**from.theta** `function(x) x`

**theta4**

**hyperid** `56604`
**name** `beta4`
**short.name** `beta4`
**output.name** `beta4 for occupancy observations`
**output.name.intern** `beta4 for occupancy observations`
**initial** `0`
**fixed** `FALSE`
**prior** `normal`
**param** `0 10`
**to.theta** `function(x) x`
**from.theta** `function(x) x`

**theta5**

**hyperid** `56605`
**name** `beta5`
**short.name** `beta5`
**output.name** `beta5 for occupancy observations`
**output.name.intern** `beta5 for occupancy observations`
**initial** `0`
**fixed** `FALSE`
**prior** `normal`
**param** `0 10`
**to.theta** `function(x) x`
**from.theta** `function(x) x`

**theta6**

**hyperid** `56606`
**name** `beta6`
**short.name** `beta6`
**output.name** `beta6 for occupancy observations`
**output.name.intern** `beta6 for occupancy observations`
**initial** `0`
**fixed** `FALSE`
**prior** `normal`
**param** `0 10`
**to.theta** `function(x) x`
**from.theta** `function(x) x`

**theta7**

**hyperid** 56607
**name** beta7
**short.name** beta7
**output.name** beta7 for occupancy observations
**output.name.intern** beta7 for occupancy observations
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 10
**to.theta** function(x) x
**from.theta** function(x) x

**theta8**

**hyperid** 56608
**name** beta8
**short.name** beta8
**output.name** beta8 for occupancy observations
**output.name.intern** beta8 for occupancy observations
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 10
**to.theta** function(x) x
**from.theta** function(x) x

**theta9**

**hyperid** 56609
**name** beta9
**short.name** beta9
**output.name** beta9 for occupancy observations
**output.name.intern** beta9 for occupancy observations
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 10
**to.theta** function(x) x
**from.theta** function(x) x

**theta10**

**hyperid** 56610
**name** beta10
**short.name** beta10
**output.name** beta10 for occupancy observations
**output.name.intern** beta10 for occupancy observations
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 10
**to.theta** function(x) x
**from.theta** function(x) x

**survival** FALSE

**discrete** `TRUE`

**link** `default logit cloglog`

**link.simple** `default logit cloglog`

**pdf** `occupancy`

### Example

```
n <- 1000
m <- 3
nc <- 2
beta <- c(-1, rnorm(nc-1, sd = 0.2))
Y <- matrix(NA, n, m)
X <- matrix(NA, n, m*nc)
z <- rnorm(n, mean = 0, sd = 0.3)
eta <- 3 + z
p.obs <- 1/(1 + exp(-eta))
for (i in 1:n) {
    is.zero <- rbinom(1, size = 1, prob = 1 - p.obs[i])
    nyy <- sample(2:m, 1)
    for(j in 1:m) {
        off <- (j-1) * nc
        if (j <= nyy) {
            X[i, off + 1:nc] <- c(1, rnorm(nc-1, sd = 0.1))
            eeta <- sum(X[i, off + 1:nc] * beta)
            p <- 1/(1 + exp(-eeta))
            if (is.zero) {
                Y[i, j] <- 0
            } else {
                Y[i, j] <- rbinom(1, size = 1, prob = p)
            }
        } else {
            X[i, off + 1:nc] <- rep(NA, nc)
            Y[i, j] <- NA
        }
    }
}

r <- inla(inla.mdata(Y, X) ~ 1 + z,
          family = "occupancy",
          data = list(Y = Y, X = X, z = z),
          control.fixed = list(prec.intercept = 1),
          control.inla = list(cmin = 0.0))
summary(r)
cbind(beta, r$summary.hyperpar[, "mean"])
```